MSc in Computer Science and Engineering

Planning, Learning and Decision Making

Lab 1: Markov Chains
Smart Home Energy Management

1 System Specification

You are tasked with modeling a Smart Home Energy System as a discrete-state Markov Chain. The system must be implemented according to the following factorization to ensure consistency across all student submissions. The questions might accept more than one correct solution, justify your options.

The state is defined by the tuple (B, W, C), where:

- **Battery Level (B):** $\{0, 1, 2, 3, 4\}$ (5 levels).
- **Weather Situation (W):** $\{0, 1, 2, 3, 4\}$ (5 levels: 0=Stormy, 4=Sunny).
- **Consumption Level (C):** $\{0, 1, 2\}$ (3 levels: 0=Low, 2=High).

Total operational states: $5 \times 5 \times 3 = 75$.
State 75 (Failure): A terminal "sink" state representing a permanent system failure.

1.1 Index Mapping

To represent the transition matrix $P \in \mathbb{R}^{76\times76}$, use the following lexicographical indexing:

$$Index = (B \times 15) + (W \times 3) + C$$

The failure state is assigned $Index = 75$.
Transitions occur in two phases: the physical update and the environmental shift.

1. **Battery Update:** The next potential battery state B' is calculated as:

$$B' = \text{clip}(B + W - C, 0, 4)$$

2. **Environmental Markov Chains:**
   - **Weather ($W \to W_{next}$):** Prob. 0.6 to stay, 0.2 to move to $W \pm 1$ (bounded by [0,4]).
   - **Consumption ($C \to C_{next}$):** Prob. 0.7 to stay, 0.15 to move to $C \pm 1$ (bounded by [0,2]).

1.2 Scenarios

You must implement three versions of the transition matrix P:

- **Scenario A (Robust)**: No transitions to State 75.
- **Scenario B (Brittle-Bottom)**: There is a small probability of battery failure. We can implement this as follows: If the transition probability to state 0 is above 10%, remove exactly 10% from this probability and add it to the transition probability of state 75.
- **Scenario C (Extreme-Stress)**: The battery is getting old and the probability of failure increases either when full or when empty. We can implement this as follows: For every next state where the probability is above 5% and either $B' = 0$ or $B' = 4$, remove exactly 5% from this probability and add it to the transition probability of state 75.

2 Questions (answers to be submitted)

1. What is a basic method to test if the transition matrix is wrong?
   Suggestion: Implement a function `build_matrix(scenario)` that returns the 76×76 transition matrix. Where scenario is either 0 (Scenario A), 1 (Scenario B), or 2 (Scenario C).

2. Compare the theoretical probability of failure $\mu_t[75]$ (calculated via P^t) with the empirical fraction of failed simulations at each time step $t \in [0, 40]$. Explain how you did it and plot the results.
   Suggestion: Write a function that generates Nr trajectories of Ns steps starting in an initial state s_0.

3. How can you compute the stationary distribution of a Markov chain? All the methods give the same results? In which conditions?
   Suggestion: Given an initial state $s_0 = (B : 2, W : 2, C : 1)$, create two functions. One that computes the distribution μ_N after N steps, for instance using **Power Iteration**. One that computes the stationary distribution if it exists using **eigen-decomposition**.

4. In Scenario C, the stationary distribution is mathematically trivial (100% failure). Explain why "Expected Time to Failure" is a more useful metric here.
   Suggestion: Compute the expect time to transition between states.

5. How does increasing the volatility of the Weather Markov Chain (e.g., lower probability of staying in the same state) affect the probability of failure in Scenario B?
   Suggestion: Try different values in the transition matrix and evaluate the stationary distribution.

6. Based on your results, if a homeowner wants to avoid State 75 in Scenario C, should they aim for a battery level of 2 or 4?
   Suggestion: Justify using your transition matrix.

3 How to submit

You should submit one file called *padiproj1-groupXX.zip* with the 3 files:

1. *padihw1-groupXX.pdf* with the homework
2. *padilab1-groupXX.pdf* with the answers to the questions in this lab
3. *padilab1-groupXX.py* with the code developed

No other formats will be accepted. The homework and the answers to the lab questions can be either a photo of handwritten notes or in a digital format, just ensure that it is readable.